Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2026

Course: B.Tech | *Time: 03 Hours*
Code: 100816 | *Subject: Bitcoin and Crypto Currencies* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There areNINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The output size of SHA-256 is
(i) 64 bits (iii) 256 bits
(ii) 128 bits (iv) 512 bits

(b) Which technique is used to convert plain text into unreadable text?
(i) Decryption (iii) Hashing
(ii) Encryption (iv) Compression

(c) Which data structure is used for efficient verification in blockchain?
(i) Binary Tree (iii) B Tree
(ii) AVL Tree (iv) Merkle Tree

(d) ECDSA stands for
(i) Elliptic Curve Digital Signature Algorithm (iii) Elliptic Curve Data Security Application
(ii) Electronic Curve Data Signature Algorithm (iv) Electronic Cryptographic Signature Algorithm

(e) Which key is used to create a digital signature?
(i) Public key (iii) Session key
(ii) Private key (iv) Shared key

(f) Which digital signature algorithm uses in Bitcoin?
(i) RSA (iii) ECDSA
(ii) DES (iv) AES

(g) Blockchain technology supports
(i) Decentralization (iii) Isolation
(ii) Centralization (iv) Duplication

(h) Which mechanism is used in Bitcoin consensus?
(i) Captcha (iii) Compression
(ii) Password checking (iv) Proof of Work

(i) The first block in blockchain is called
(i) Child block (iii) Genesis block
(ii) Root block (iv) Parent block

(j) Which wallet type is safer for long-term storage?
(i) Hot wallet (iii) Online wallet
(ii) Cold wallet (iv) Mobile wallet

Q.2 (a) How does SHA-256 help in maintaining blockchain security and immutability? **[7]**
(b) Explain the working of a Merkle tree and discuss why it is preferred in large distributed systems. **[7]**

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Q.3	(a)	Why digital signatures are considered more reliable than handwritten signatures in electronic transactions?	[7]
	(b)	Compare RSA and ECDSA algorithms.	[7]
Q.4	(a)	Why is decentralization considered more secure than centralized systems in blockchain technology?	[7]
	(b)	Explain how distributed consensus solves the double-spending problem in cryptocurrencies.	[7]
Q.5	(a)	Describe the role of miners and nodes in validating Bitcoin transactions.	[7]
	(b)	Discuss possible security risks in Bitcoin transaction scripts.	[7]
Q.6	(a)	Write down the differences between hot wallets and cold wallets.	[7]
	(b)	Explain how splitting and sharing keys improves fault tolerance and security.	[7]
Q.7	(a)	Explain possible ethical and legal challenges associated with anonymous cryptocurrencies.	[7]
	(b)	Discuss the privacy advantages and limitations of Zerocoin and Zerocash.	[7]
Q.8	(a)	Define cryptography. Explain the basic concepts and goals of cryptography.	[7]
	(b)	Discuss applications of digital signatures in blockchain technology.	[7]
Q.9	(a)	Explain the Genesis block and blockchain linking process.	[7]
	(b)	Discuss the advantages and disadvantages of centralized exchanges.	[7]

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